

1. Rocking/jittering is probably due to artifacts within the solver. Since the contact point manifold has multiple points, and the engine is not setup for parallelized touch point calculations, there's a small jitter as the engine resolves each downward acceleration vector of the box. So there's a mismatch in the cancellation of the normal and tangential vectors, the residual vector is what gives us the jitter.

mu	Final tangential speed	Average tangential acceleration over last half	Interpretation
0.12	9.669751	2.120449	Strong sliding and clear acceleration
0.18	5.255719	1.219703	Still accelerates down the ramp
0.22	0.785928	0.105896	Below $\tan(\theta)$, so it continues to accelerate
0.24	0.082877	0.003215	Above threshold, but the engine still creeps
0.30	0.052580	-0.001510	Nearly static, with residual numerical drift
0.50	0.030827	0.001717	Very small creep only
0.80	-0.025109	0.001432	Tiny oscillatory drift

2.
 - a. Generated this table programmatically and field out the interpretation to show the effect of mu on the motion of the block.
3. Adding a tangential drift correction vector seemed to help a lot, especially if the contact is within the static friction domain. Beyond that threshold, where kinetic friction dominates, the magnitude of this correction is turned way down, so the block can still slide freely, then when the block is at rest, the vectors are zeroed out and gravity is no longer applied, so the body is fully at rest.

Racer	Measured final speed	Analytic speed	Percent error
Sliding particle	5.190916	5.195587	-0.089906%
Sphere	4.384223	4.391073	-0.155991%
Solid cylinder	4.234995	4.242179	-0.169343%
Hollow cylinder	3.665601	3.673835	-0.224111%

4.

- a. These match expectations, The slider is fastest because no energy is stored in rotation. Among the rolling cases, a smaller rotational inertia gives a larger translational speed, so the sphere beats the solid cylinder, and the solid cylinder beats the hollow cylinder.